F12 4th Carlou

Fall 2012

MAC 1147 - Precalculus Algebra/Trig

INSTRUCTOR: Sherry Tornwall

OFFICE: Little 374

OFFICE HOURS: M5, T4, W3, R6, F5

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* COURSE WEBSITE: E-Learning Sakai, http://lss.at.ufl.edu

TEXTBOOK: Precalculus with Limits, 2nd edition, by Larson

ONLINE HOMEWORK: http://www.webassign.net

EVENING EXAMS: 8:30-10 pm, See Sakai for dates.

FINAL EXAM: 12/8/12 Saturday, 3:00 pm

COURSE GRADE

Course Intro Quizzes and Self-Eval on Sakai (5 pts)

Book homework (3 @ 10 pts for max of 20pts)

Discussion quizzes (best 8 out of 10 @ 10 pts)

Online WebAssign HW (13 @ 3pts for max of 30 pts)

Online WebAssign quizzes (best 10 out of 13 @ 4 pts)

Exams best 3 of 4 @ 75pts

Cumulative Final (100 pts)

No Calculators are allowed for any of the quizzes and exams.

Check your grade at E-Learning Sakai, https://lss.at.ufl.edu/ at least once a week. Disputed points will NOT be reviewed at the end of semester.

not real - complex # $\sqrt{-9}$

Lecture 1: Section A.1
Real Numbers and Their Properties

Real Numbers:

Real Numbers R

Rational Numbers Q	Irrational Numbers I
$Q = \left\{ \frac{p}{q} \middle| p \text{ and } q \text{ integers, } q \neq 0 \right\}$	$I = \{x \mid x \text{ not rationals}\}$
Integers Z $Z = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$	

ex. Rational numbers:

1) fractions and integers: $\frac{1}{5}$, $-\frac{2}{7}$, 0, -3, $\frac{2}{3} = 0.\overline{6}$

2) finite decimals: 0.15, -2.18

$= \frac{2}{9}$

3) infinite repeating decimals: $0.\overline{2}$, $-1.2\overline{75}$

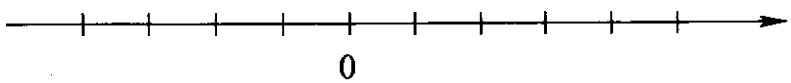
ex. Irrational numbers:

infinite nonrepeating decimals: $\sqrt{2}$, $\sqrt[3]{4}$, π, e,

$1.010010001\ldots$

Real Number Line

Every real number can be represented graphically as a point on the real number line.

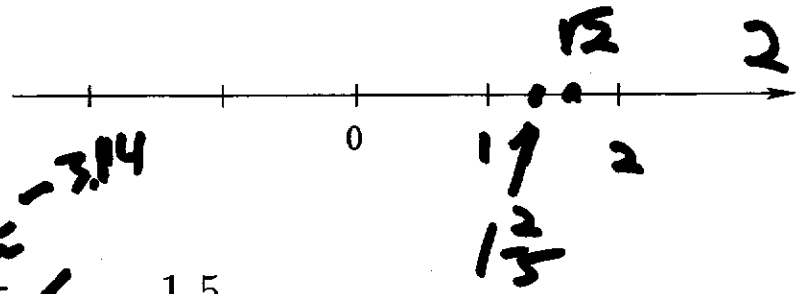


ex. Order the numbers by placing the appropriate inequality symbol ($<$ or $>$) between the pair of real numbers. Plot them on the real number line.

1) $\sqrt{2} > \frac{7}{5} = 1\frac{2}{5}$

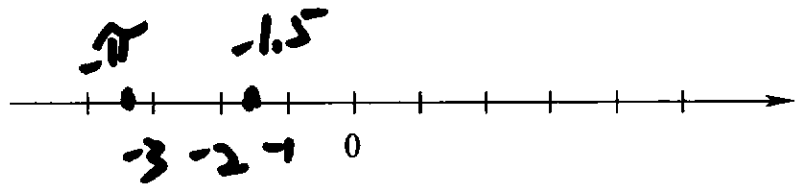
$(\sqrt{2})^2 \quad \left(\frac{7}{5}\right)^2$

$2 > \frac{49}{25}$



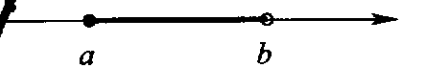
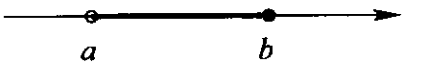
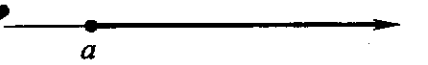
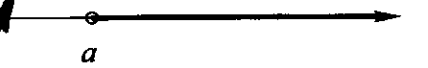
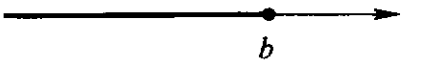
2) $-\pi < -1.5$

≈ -3.14



Closed interval iff $[a,b]$
open interval iff (a,b)
neither, not either of the above
Bounded interval – has a finite length.

Intervals

	Interval Notation	Inequality Notation	Graph
closed	$[a,b]$	$a \le x \le b$	
open	(a,b)	$a < x < b$	
neither	$[a,b)$	$a \le x < b$	
neither	$(a,b]$	$a < x \le b$	
neither	$[a,\infty)$	$x \ge a$	
open	(a,∞)	$x > a$	
neither	$(-\infty,b]$	$x \le b$	
open	$(-\infty,b)$	$x < b$	
open	$(-\infty,\infty)$	$-\infty < x < \infty$	

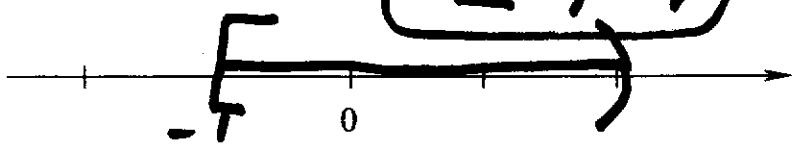
bounded

unbounded

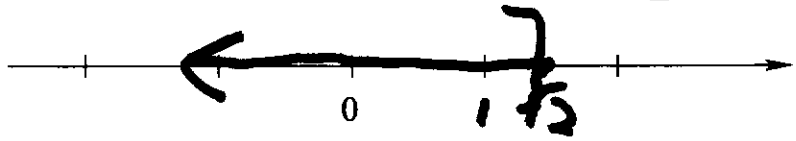
NOTE: The symbols ∞ (positive infinity) and $-\infty$ (negative infinity) do not represent real numbers.

ex. Express each of the following as intervals or inequalities, and then graph each on the number line.

1) $-1 \leq x < 2$ $\boxed{[-1, 2)}$

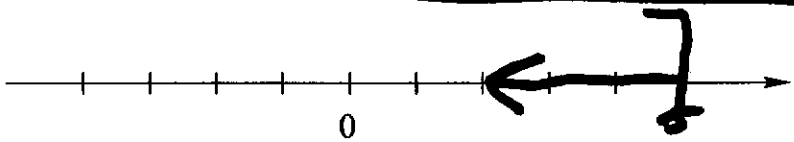


2) $(-\infty, \sqrt{2}]$ $\boxed{x \leq \sqrt{2}}$

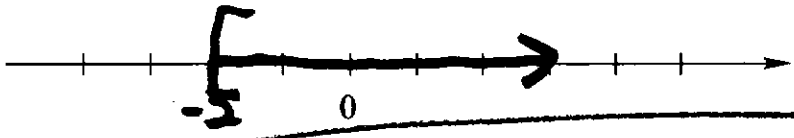


ex. Use interval and inequality notations to describe the following.

1) x is at most 5. $\boxed{(-\infty, 5] \quad x \leq 5}$



2) y is at least -2.

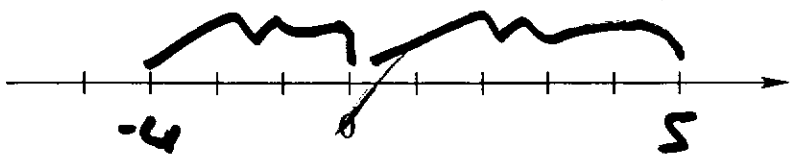


$[-2, \infty) \quad y \geq -2$

Absolute Value

Def. The **absolute value** of a real number a is its magnitude, or the distance between the origin and the number a on the real number line.

$|-4| = 4$ $\qquad$ $|5| = 5$



* $|a| = \begin{cases} a, & \text{if } a \geq 0 \\ -a, & \text{if } a < 0 \end{cases}$ is the distance between the origin

$|-4| = -(-4) = 4$

and the number a.

ex. Write the expression without absolute value bars.

neg

1) $|3 - \pi| = -(3-\pi) = \boxed{-3+\pi = \pi - 3}$

neg

2) $-|1 - \sqrt{2}| =$

Let $x = -3$

$-(-(1-\sqrt{2})) = \boxed{1-\sqrt{2}}$

$\frac{|-3|}{-3} = \frac{3}{-3} = -1$

3) If $x \neq 0$, $\frac{|x|}{x} =$ if $x > 0$, then $\frac{x}{x} = 1$

if $x < 0$, then $\frac{-x}{x} = -1$

Distance on the Number Line

The distance between points a and b on the number line is

$$d(a,b) = |b-a| = |a-b|$$

<u>ex.</u> Find the distance between -1 and -8 on the number line.

$d(-1,-8) = |-8-(-1)| = |-8+1| = |-7| = 7$

$= |-1-(-8)| = |-1+8| = |7| = 7$

<u>ex.</u> Write using the absolute value: The distance between x and -2 is at least 5.

$|x-(-2)| \geq 5 \rightarrow |x+2| \geq 5$

Properties of Absolute Values

For any real numbers a and b,

1. $|a| \geq 0$
2. $|-a| = |a|$
3. $|ab| = |a||b|$
4. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, $b \neq 0$

9

<u>ex.</u> Simplify the expression $-|-5| - |-9|$

$-5-9 = -14$

Algebraic Expressions

Def. An **algebraic expression** is a collection of letters (variables) and real numbers (constants) combined using the operations of $+$, $-$, $\times$, $\div$, and exponentiation.

variable terms

constant

4 terms

coefficient

ex. $2x^2 - 4x + 1 + \frac{1}{x}$

2 terms

ex. $\frac{5x^2 + y}{2z} = \frac{5x^2}{2z} + \frac{y}{2z}$

Def. The **terms** of an algebra expression are those parts that are separated by addition. Those containing a variable are called variable terms, the other is called constant term. The numerical factor of a variable term is the **coefficient** of the variable term.

ex. Identify the terms. Then identify the coefficients of the variable terms of the expression $3\sqrt{2}x^3 - 4x + 2$.

3 terms

$3\sqrt{2}x^3$

coef.

$-4x$

variable terms

2 constant term

Read

Basic Rules of Algebra

Let a, b, and c be real numbers, variables, or algebraic expressions.

Commutative Properties:

$a + b = b + a$ $\qquad$ $ab = ba$

Associative Properties:

$(a + b) + c = a + (b + c)$ $\qquad$ $(ab)c = a(bc)$

Distributive Properties:

$a(b + c) = ab + ac$ $\qquad$ $(a + b)c = ac + bc$

Identity Properties:

$a + 0 = a$ (0 is the additive identity)

$a \cdot 1 = a$ (1 is the multiplicative identity)

Inverse Properties:

$a + (-a) = 0$ ($-a$ is the additive inverse of a)

$a \cdot \frac{1}{a} = 1,\ a \neq 0$ ($\frac{1}{a}$ is the multiplicative inverse of a)

Read

Let a and b be real numbers, variables, or algebraic expressions.

Properties of Negation

1. $(-1)a = -a$
2. $-(-a) = a$
3. $(-a)b = -(ab) = a(-b)$
4. $(-a)(-b) = ab$
5. $-(a+b) = (-a) + (-b)$

Properties of Equality

1. If $a = b$, then $a \pm c = b \pm c$ and $ac = bc$.
2. If $a \pm c = b \pm c$, then $a = b$.
3. If $ac = bc$ and $c \neq 0$, then $a = b$

Properties of Zero

1. $a \pm 0 = a$
2. $a \cdot 0 = 0$
3. $\dfrac{0}{a} = 0, \ a \neq 0$
4. $\dfrac{a}{0}$ is undefined.
5. <u>Zero-Factor Property:</u>

   If $ab = 0$, then either $a = 0$ or $b = 0$

Read

Properties and Operations of Fractions

Let a, b , c and d be real numbers, variables, or algebraic expressions with $b \neq 0$ and $d \neq 0$.

1. Equivalent Fractions: $\dfrac{a}{b} = \dfrac{c}{d} \Leftrightarrow ad = bc.$

2. Rule of Signs: $-\dfrac{a}{b} = \dfrac{-a}{b} = \dfrac{a}{-b}$ and $\dfrac{-a}{-b} = \dfrac{a}{b}$

3. Generate Equivalent Fractions: $\dfrac{a}{b} = \dfrac{ac}{bc}, \ c \neq 0$

4. Add or Subtract with Like Denominators:
$$\frac{a}{b} \pm \frac{c}{b} = \frac{a \pm c}{b}$$

5. Add or Subtract with Unlike Denominators:
$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

6. Multiply Fractions: $\dfrac{a}{b} \cdot \dfrac{c}{d} = \dfrac{ac}{bd}$

7. Divide Fractions: $\dfrac{a}{b} \div \dfrac{c}{d} = \dfrac{a}{b} \cdot \dfrac{d}{c} = \dfrac{ad}{bc}$

ex. $\left(\frac{1}{9}-\frac{1}{8}\right)\div\frac{1}{72}=$

$\left(\left(\frac{8}{8}\right)\frac{1}{9}-\frac{1}{8}\left(\frac{9}{9}\right)\right)\div\frac{1}{72}$ $=\left(\frac{8}{72}-\frac{9}{72}\right)\div\frac{1}{72}$

$\frac{-1}{\cancel{72}}\times\frac{\cancel{72}}{1}=-1$

ex. $\left(2\div\frac{2}{3}\right)-\left(\frac{2}{3}\div 2\right)=$

$\left(\frac{2}{1}\div\frac{2}{3}\right)-\left(\frac{2}{3}\div\frac{2}{1}\right)=$

$\left(\frac{\cancel{2}}{1}\times\frac{3}{\cancel{2}}\right)-\left(\frac{\cancel{2}}{3}\times\frac{1}{\cancel{2}}\right)=3-\frac{1}{3}=\frac{9}{3}-\frac{1}{3}=\frac{8}{3}$

ex. $\dfrac{-\frac{9}{25}}{-\frac{36}{15}}=-\frac{9}{25}\div-\frac{36}{15}=$

$-\frac{\cancel{9}}{\cancel{25}_{5}}\times-\frac{\cancel{15}^{3}}{\cancel{36}_{4}}=+\frac{1}{5}\times\frac{3}{4}=\boxed{\frac{3}{20}}$

Practice.

1) Let $A = \left\{3,\ \pi,\ \frac{3}{4},\ 0.\overline{3},\ -7,\ -\frac{2}{3},\ \sqrt{3},\ 0.271357\ldots\right\}$.
What numbers in set A are rational numbers?

2) Express the interval $(-3, \infty)$ in terms of an inequality and then graph the interval.

3) Use interval and inequality notation to describe:

z is less than 2 but no less than -4.

4) Evaluate

$356 \times 0.23 - 523 \times 0.92 + 644 \times 0.23 - 477 \times 0.92$

Answer. 1) $3, \frac{3}{4}, 0.\overline{3}, -7, -\frac{2}{3}$ 2) $x > -3$ 3) $-4 \le z < 2$ 4) -690